Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Need for statistics: purposes, and the collection and tabulation of data. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	9
Topic	Need for statistic
Subtopic	Need for statistics: purposes, and the collection and tabulation of data
Theme	—
Curriculum objectives	• explain the purpose of statistic • analyse and interpret statistical data presented in tables
Previous knowledge	• Students are familiar with counting and recording information.
Lesson sequence	• explain the purpose of statistic — the teacher introduces the concept of statistics as a method of collecting and organising information • analyse and interpret statistical data presented in tables — the teacher takes the students through collecting data, recording it with tally marks, and setting it out in a frequency table
Instructional materials	• classroom register, bottle tops, pencils or counters.
Evaluation	• (a) Define statistic • (b) State three methods of collecting data. • (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Lesson objectives

1.	explain the purpose of statistic
2.	analyse and interpret statistical data presented in tables

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Need for statistics: purposes, and the collection and tabulation of data

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: Why We Use Statistics

Every day, you come across lots of facts and figures — how many goals a team scored, the temperature outside, the price of bread, your scores in a test. When these pieces of information are gathered together, we call them data. Statistics is the part of mathematics that deals with collecting data, organising it, and making sense of it.

Why do we need statistics? Its main purpose is to help us make decisions based on facts instead of guesswork. For example, a shopkeeper who records which snacks sell best each week can decide which ones to stock more of. A school that records how many students are absent each day can see when illnesses spread and take action. A football coach who notes how many goals each player scores can pick the strongest team.

Statistics does three main jobs:

Collecting data — gathering facts from a source, such as a survey or a tally.
Organising data — putting the facts in order so they are easy to read, often in a table.
Interpreting data — reading the organised facts to answer questions and draw conclusions.

In this section you will learn the purpose of statistics and how to read the information that is laid out in a table.

Worked example 1 — reading a table

A class recorded how each student travelled to school one morning. The results were put into a table.

<table>
<tr><th>Way of travelling</th><th>Number of students</th></tr>
<tr><td>Walking</td><td>14</td></tr>
<tr><td>Bicycle</td><td>8</td></tr>
<tr><td>Bus</td><td>6</td></tr>
<tr><td>Car</td><td>4</td></tr>
</table>

Question: Which way of travelling was used by the most students?

Answer: Walking. The table shows 14 students walked, and no other row has a bigger number, so walking was the most common way.

The table lets us answer this question at a glance. If the same facts were just a long jumbled list of words, we would have to count through everything again. Organising the data in a table is what makes the information useful.

The purpose of statistics is not just to count things — it is to use the counts to decide and predict. When data is set out in a table, you can quickly find the biggest value, the smallest value, the total, and the difference between values. These simple readings let you answer real questions.

The steps you follow are:

Read the headings of the table to know what each column and row mean.
Compare the numbers to answer the question — look for the largest, the smallest, or the total.
State your conclusion in a full sentence that uses the data you read.

Remember that a conclusion is only trustworthy when it comes from the data shown. If a number is not in the table, you cannot invent it.

Worked example 2 — comparing values in a table

A fruit seller recorded how many of each fruit he sold on Saturday.

<table>
<tr><th>Fruit</th><th>Fruits sold</th></tr>
<tr><td>Apples</td><td>25</td></tr>
<tr><td>Oranges</td><td>30</td></tr>
<tr><td>Bananas</td><td>18</td></tr>
<tr><td>Mangoes</td><td>12</td></tr>
</table>

Question 1: How many more oranges than mangoes were sold?

Answer 1: 30 − 12 = 18, so 18 more oranges were sold.

Question 2: How many fruits were sold altogether?

Answer 2: 25 + 30 + 18 + 12 = 85 fruits.

Question 3: Which fruit should the seller order more of next week?

Answer 3: Oranges, because they had the largest number sold (30). The table tells him which fruit people wanted most.

Practice

1. A shop recorded the number of drinks sold on Friday.

Drink	Drinks sold
Water	40
Juice	25
Soda	30
Tea	15

   (a) Which drink sold the most?
   (b) How many drinks were sold altogether?
2. A teacher recorded the number of students in each house for a sports competition.

House	Number of students
Red	45
Blue	55
Green	50

   (a) Which house has the fewest students?
   (b) How many more students are in Blue than in Red?
   (c) If the teacher wants to reward the house with the most students, which house should she reward?

Section 2: Reading Stories from Tables of Data

When a survey or count is finished, the results are often long lists of numbers that are hard to take in at a glance. A table gathers those results into neat rows and columns so that we can see patterns, compare values, and answer questions quickly. Tables turn a messy pile of facts into a clear picture.

To read a table, look first at the heading, which tells you what was counted. Then read the row labels down the left and the column labels across the top. Where a row and a column meet, the number in that box is the answer to the question the two labels ask together.

Here is an example. The table below shows the favourite sport chosen by students in one school, with the number of students who chose each sport.

Favourite sport	Number of students
Football	40
Basketball	25
Swimming	20
Athletics	15
Total	100

Three questions you can answer from this table:

How many students chose swimming? Look along the "Swimming" row to the number column: 20.
Which sport was most popular? The row with the biggest number is Football (40), so football was the most popular.
How many students took part altogether? The Total row at the bottom says 100.

A "Total" row or column is a great help: it lets you check that all the parts add up, and it gives you a whole to compare each part against.

The word statistic is about collecting, organising and making sense of facts in numbers. Tables are one of the clearest ways to organise data. When you analyse a table, you look carefully at the numbers to spot what is most common, what is least common, and how the parts compare. When you interpret it, you explain in words what the numbers mean.

A good way to work through a table is to answer questions in this order:

What does the table count? Read the heading and the labels.
How much is each part? Read the number in each row or column.
What is the total? Add the parts or read the total row.
What is the pattern? Compare the parts to find the biggest, the smallest, and how they differ.

Following these steps turns a table of numbers into a sentence you can say aloud, like "Twice as many students walk to school as come by bicycle."

Worked example

The table below shows how students in a class travel to school each day.

How students travel	Number of students
Walk	30
Bicycle	15
Bus	10
Car	5
Total	60

Let's analyse and interpret it step by step.

Step 1 — What does the table count?
The heading tells us the number of students using each way of travelling to school.

Step 2 — How much is each part?
Walk: 30, Bicycle: 15, Bus: 10, Car: 5.

Step 3 — What is the total?
The Total row reads 60. Check: 30 + 15 + 10 + 5 = 60. ✓

Step 4 — What is the pattern?
Walking is the most common way (30), and travelling by car is the least common (5). Walking is used by twice as many students as bicycling (30 = 2 × 15).

Interpretation in words: Most students in the class walk to school, and very few come by car.

Practice

1. The table below shows how many books students in a library borrowed in one week.

Type of book	Number borrowed
Story books	45
Science books	20
History books	15
Poetry books	10
Total	90

   (a) How many science books were borrowed?
   (b) Which type of book was borrowed the most?
   (c) Which type of book was borrowed the least?
   (d) How many books were borrowed altogether?
   (e) Write one sentence interpreting what the table tells you.

2. The table below shows the number of pupils in a school who chose each school club.

School club	Number of pupils
Drama	35
Music	30
Art	25
Chess	10
Total	100

   (a) Which club was chosen by the fewest pupils?
   (b) How many more pupils chose Drama than Music?
   (c) How many pupils chose Art or Chess together?
   (d) What fraction of the whole school chose Music?

Check your work

Attempt every question before you read this page.

Why We Use Statistics

1. (a) Water, because 40 is the largest number in the table. (b) 40 + 25 + 30 + 15 = 110 drinks.

   (a) Look down the "Drinks sold" column and find the biggest number. The numbers are 40, 25, 30 and 15; the biggest is 40, which belongs to Water.
   (b) To find the total, add all four numbers: 40 + 25 = 65, 65 + 30 = 95, 95 + 15 = 110. So 110 drinks were sold altogether.

2. (a) Red, because 45 is the smallest number. (b) 55 − 45 = 10 more students. (c) Blue, because it has the most students (55).

   (a) Compare the numbers 45, 55 and 50. The smallest is 45, which is Red.
   (b) Subtract the smaller from the larger: 55 − 45 = 10. So Blue has 10 more students than Red.
   (c) The house with the most students is the one with the largest number, 55, which is Blue. The table lets the teacher decide using the facts instead of guesswork.

Reading Stories from Tables of Data

1. (a) 20 science books. (b) Story books (45, the biggest number). (c) Poetry books (10, the smallest number). (d) 90 books altogether. (e) Example: Most books borrowed were story books, and poetry books were borrowed the least.

   (a) Look along the "Science books" row to the number column: 20. (b) The row with the largest number is Story books at 45, so story books were borrowed most. (c) The row with the smallest number is Poetry books at 10. (d) The Total row reads 90; check $45 + 20 + 15 + 10 = 90$. (e) Compare the parts: the largest is story books and the smallest is poetry books, so we can say most borrowed were story books and fewest were poetry books.
2. (a) Chess (10 pupils). (b) 5 more pupils chose Drama ($35 - 30 = 5$). (c) 35 pupils chose Art or Chess together ($25 + 10 = 35$). (d) 30 out of 100, which is 3/10 (thirty hundredths).

   (a) The smallest number in the table is 10, which belongs to Chess. (b) Drama has 35 and Music has 30; the difference is $35 - 30 = 5$. (c) Add the Art and Chess rows: $25 + 10 = 35$. (d) Music has 30 pupils out of a total of 100, so the fraction is $30/100 = 3/10$.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-0a383bae	**Date**	
Subtopic	Need for statistics: purposes, and the collection and tabulation of data	**Rater**	

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____